



Learning Residual-Free Bubbles

to Solve Elliptic Problems

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GA036 – Introdução ao Aprendizado de Máquina

1. The model

We consider the one-dimensional reaction-advection-diffusion problem

$$-(\varepsilon u')' + \beta u' + \sigma u = f \quad \text{in } (0, \ell), \quad u(0) = u(\ell) = 0.$$

Weak form. Find $u \in H_0^1(0, \ell)$ such that

$$a(u, v) = L(v) \quad \forall v \in H_0^1(0, \ell),$$

with

$$a(w, v) = \int_0^\ell \varepsilon w' v' + \beta w' v + \sigma w v \, dx,$$

$$L(v) = \int_0^\ell f v \, dx.$$

2. The Galerkin idea

Let $V_h^1 \subset H_0^1(0, \ell)$ be the continuous piecewise-linear finite element space.

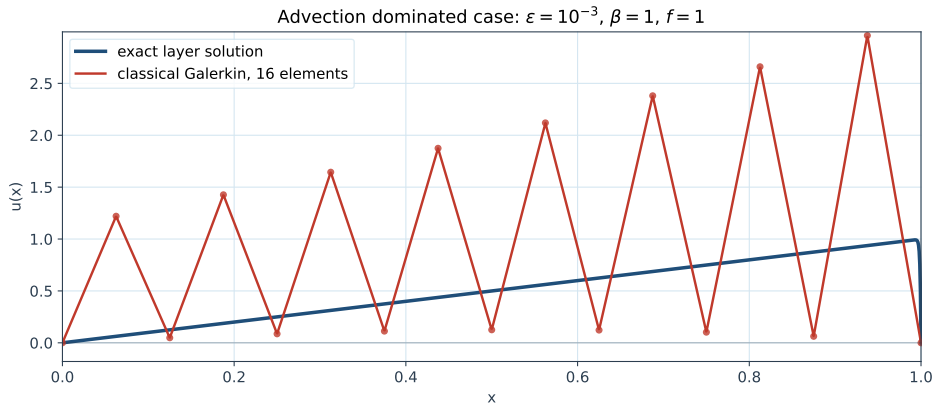
$$V_h^1 = \{v_1 \in H_0^1(0, \ell) : v_1|_K \in \mathbb{P}_1(K)\}.$$

The classical method searches for $u_1 \in V_h^1$ such that

$$a(u_1, v_1) = L(v_1) \quad \forall v_1 \in V_h^1.$$

Galerkin methods are attractive because they produce sparse linear systems, respect variational structure, and handle boundary conditions naturally. The difficulty is that V_h^1 may be too poor to capture sharp layers, oscillating when the element Péclet number $\text{Pe} = \beta h / (2\varepsilon) \gg 1$.

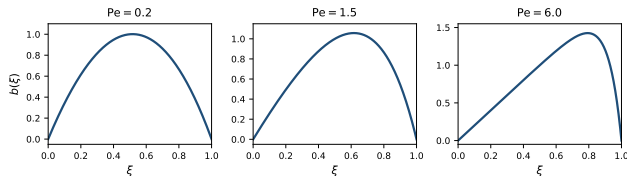
3. How can we *correct* $u_1 \in V_h^1$?



u_1 is an unstable approximation. We want some correction $u_B \in H_0^1(0, \ell)$ such that $u_1 + u_B$ is a better approximation.

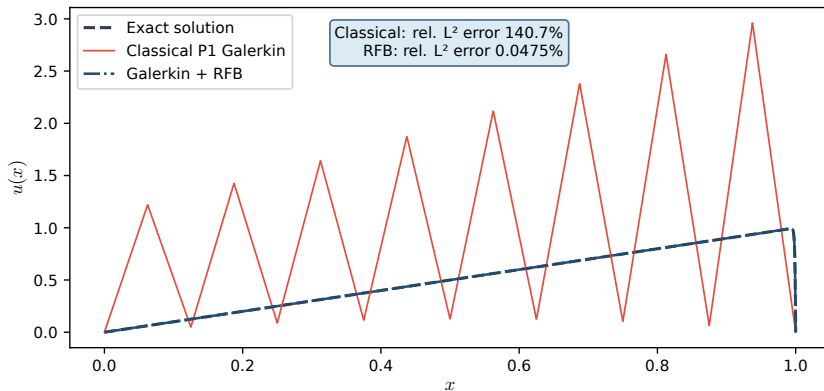
4. Residual-free bubbles

For a coarse approximation $u_1 \in V_h^1$, define the element residual $r_K(u_1) = f - \mathcal{L}u_1$ with $\mathcal{L}w = -(\varepsilon w')' + \beta w' + \sigma w$. The ideal bubble correction on K solves $\mathcal{L}(b_K) = r_K(u_1)$ in K with $(b_K)|_{\partial K} = 0$.



By construction $\mathcal{L}(u_1 + b_K) = f$ on K : the ideal bubble exactly cancels the element residual. Enriching the space V_h^1 with these bubbles and solving the global Galerkin problem yields the RFB approximation $u_h = u_1 + \sum_K b_K$ (Franca and Farhat 1995; Brezzi, Franca, and Russo 1998).

4. Residual-free bubbles



With *exact* bubbles the enriched space $V_h^1 \oplus B$ contains the exact solution when f is resolved by the bubble basis. Knowing the bubble map $u_1 \mapsto u_B$ is the key to making this approach practical without mesh refinement.

5. The idea: learn the bubble functions

Do not compute every bubble. Learn the bubble.

Offline. Solve many local RFB problems once to create training data.

Online. Evaluate the learned model from local parameters and statically condense the bubble.

Learning good b_K 's feeds the approximation $u_h = u_1 + \sum_K b_K$, thus improving the overall solution.

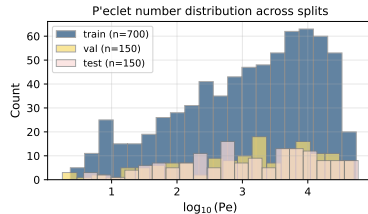
6. Data Generation ($\varepsilon, \sigma, \beta \in \mathbb{R}^+$)

We create a dataset $\mathcal{D} = \{(\mathbf{p}_i, \mathbf{y}_i)\}_{i=1}^N$ by solving the local bubble ODE on the reference element $[0, 1]$:

$$-\frac{\varepsilon}{h^2}b'' + \frac{\beta}{h}b' + \sigma b = r(\xi), \quad b(0) = b(1) = 0, \quad r \in \{1, \xi\}.$$

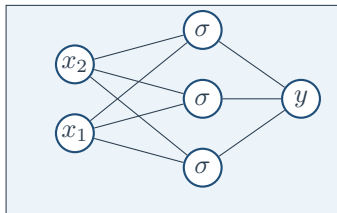
Input $\mathbf{p}_i \in \mathbb{R}^2$ $\text{Pe} = \frac{\beta h}{2\varepsilon}$, $\rho = \frac{\sigma h^2}{\varepsilon}$ from
physically relevant ranges of $(\varepsilon, \beta, \sigma)$.

Target per mode $\mathbf{y}_i \in \mathbb{R}^{2Q}$ For each mode
 $r \in \{1, \xi\}$ we store $(b(\xi_j), b'(\xi_j))_{j=1}^Q$ at Gauss
points $\xi_j \in [0, 1]$, where b solves $\mathcal{L}_K b = r$ via finite
differences.

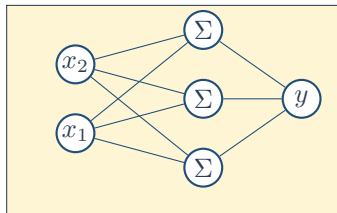


7. What is a KAN?

MLP



KAN



Key architectural difference

MLP: fixed nonlinearity on nodes, learnable linear scalars on edges.

KAN: learnable 1D functions (B-splines) on edges, summation on nodes.

Each ϕ is a B-spline curve: smooth, compactly supported, and efficiently trainable. This gives (almost) the optimal N^{-4} scaling law for cubic splines (Liu et al. 2024).

8. Learning bubbles with a KAN

We train a compact model that maps local parameters (Pe, ρ) to the bubble shape $b(\xi)$ for each mode $r \in \{1, \xi\}$, minimizing

$$\mathcal{J}(\theta) = \|b_\theta - b^{\text{RFB}}\|_{L^2(0,1)}^2 + \eta \|b'_\theta - (b^{\text{RFB}})'\|_{L^2(0,1)}^2,$$

where b^{RFB} is the reference FD solution.

What a KAN gives us

- ▶ Smooth trainable 1D functions on edges.
- ▶ Compact representation of the parameter-to-shape map.
- ▶ Interpretable local nonlinear transformations.

Why not a generic black box?

- ▶ The bubble is a structured local function.
- ▶ We want a compact model, not a huge network.
- ▶ Built-in constraints such as zero boundary values.

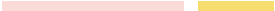
(Liu et al. 2024; Murphy 2022)

9. Advantages of the process

- ▶ Variational structure preserved, static condensation eliminates bubble DOFs locally.
- ▶ Global system size and sparsity pattern unchanged, same number of unknowns as P_1 –Galerkin, same connectivity.
- ▶ Bubble learned on the reference element with dimensionless inputs (Pe, ρ) ; h enters only through these numbers.
- ▶ Model trained once offline, reusable for any mesh h without retraining.
- ▶ Naturally adapts to heterogeneous problems: (Pe, ρ) computed per element at assembly.

References I

-  Brezzi, F., L.P. Franca, and A. Russo (1998). “Further considerations on residual-free bubbles for advective-diffusive equations”. In: *Comput. Methods Appl. Mech. Engrg.* 166.1–2, pp. 25–33. DOI: [10.1016/S0045-7825\(98\)00080-2](https://doi.org/10.1016/S0045-7825(98)00080-2).
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-  Liu, Z. et al. (2024). “KAN: Kolmogorov-Arnold Networks”. In: *arXiv:2404.19756*. arXiv: 2404.19756.
-  Murphy, K.P. (2022). *Probabilistic Machine Learning: An Introduction*. MIT Press. URL: <https://probml.github.io/pml-book/book1.html>.



Questions?

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